2nd International Conference on Industrial Artificial
Intelligence (IAI), Oct. 23-25, 2020, Shenyang, China

# Digital Twin Enabled Smart Control Engineering as an Industrial AI: A New Framework and Case Study

Jairo Viola and YangQuan Chen
*MESA Lab*
*University of California, Merced*
Merced, CA, USA
(jviola,ychen53)@ucmerced.edu

***Abstract*—In Industry 4.0, the increasing complexity of industrial systems introduces unknown dynamics that affect the performance of manufacturing processes. Thus, Digital Twin appears as a breaking technology to develop virtual representations of any complex system design, analysis, and behavior prediction tasks to enhance the system understanding via enabling capabilities like real-time analytics, or Smart Control Engineering. In this paper, a novel framework is proposed for the design and implementation of Digital Twin applications to the development of Smart Control Engineering. The framework involve the steps of system documentation, Multidomain Simulation, Behavioral Matching, and real-time monitoring, which is applied to develop the Digital Twin for a real-time vision feedback temperature uniformity control. The obtained results show that Digital Twin is a fundamental part of the transformation into Industry 4.0.**

***Index Terms*—Digital Twin, Smart Control Engineering, Behavioral Matching, Industry 4.0**

## I. Introduction

Modern manufacturing processes require high-quality standards and enhanced robustness and autonomy to achieve production objectives. So that, in Industry 4.0, the convergence of multiple breaking technologies like Deep Learning, Artificial Intelligence, Big Data, real-time analytics is required to introduce newly smart enabled capabilities into manufacturing processes. The application of these technologies in industrial environment is known as Industrial Artificial Intelligence (IAI), focused on helping an enterprise to monitor, optimize or control the behavior of these systems, improving its efficiency and performance [1]. Thus, multiple approaches of IAI can be found in the literature to deal with specific problems in the industry like predictive maintenance, supply chain, fault detection, among others [2]–[7]. In automatic control, IAI integration offers to the control engineers new tools to improve the control performance transforming the classic Control Engineering into Smart Control Engineering (SCE) [8]. Thus, a unified, detailed, and realistic representation of the smart systems is required in order to leverage the breaking technologies and IAI to enable SCE. So that, Digital Twin (DT) offers the possibility of having a highly detailed and realistic virtual model of any system that replicates its real behavior based on the system physics and data-driven models of each subsystem. In the recent years, there are many DT applications for drones, UAV [9], power systems [10], CNC machines [11], smart transportation [12], smart grid [13], [14] , or smart cities [15]. Therefore, a new framework is required to develop Digital Twin applications for SCE.

This paper presents a novel framework for developing Digital Twin applications for industrial process control applications towards a SCE. This framework consists of five steps: Target System Definition, System Documentation, Multidomain Simulation, Digital Twin assembly, Behavioral Matching, and DT Validation and Deployment. This framework is applied for designing the Digital Twin of a real-time vision feedback infrared temperature uniformity control. The Multidomain Simulation uses Simscape multiphysics to represent each domain according to its physical constitutive laws. The Behavioral Matching is performed using Genetic Algorithms (GA) to determine the unknown parameters at every single element in the DT. Finally, the DT deployment is done via a supervisory interface to assess in real-time thermal system performance with the Digital Twin. The main contribution of this paper is to introduce a new framework for developing DT applications in process control towards the development of SCE and its application into the IAI.

## II. Digital Twin and Smart Control Engineering

### A. What is Digital Twin?

Digital Twin (DT) can be defined as a precise, virtual copy of a machine or system driven by data collected from sensors in real-time that mirror almost every facet of a product, process, or service [16]. A graphical description of Digital Twin is presented in Fig.1. It is observed that DT model starts from a physical system, composed of multiple subsystems like sensors or actuators. Every single element has a virtual representation inside the Digital Twin environment, corresponding to a multi-domain simulation model built using different physics-based simulation tools that replicate the real system behavior. Likewise, the prototype can have multiple instances for a specific task like control system design, prognosis, or Fault simulation. Also, the instances can be aggregated if the specific tasks require multiple results based on the same virtual representation. Notice that the DT environment is feed with real-time data from the physical system through the Internet of Things and Edge Computing devices. The main features of a Digital Twin application are:

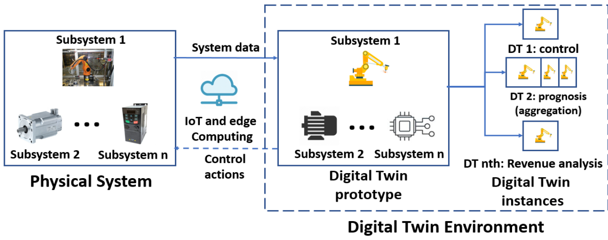


Fig. 1: Digital Twin structure

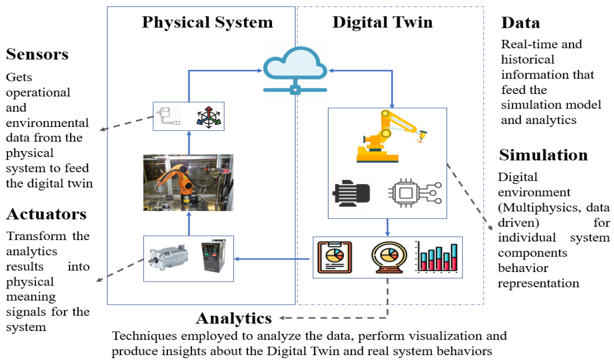


Fig. 2: Digital Twin components

- **Realizability:** The virtual representation of the physical system is performed through virtual entities or avatars.
- **Real-Time system data updating:** The sensing and analytics technologies like IoT and Edge computing enables DT real-time update.
- **Behavioral Matching:**The system behavior matching relies not only on experience-based knowledge but also in adaptive optimizing techniques feed with the real-time data updating information.
- **Modular Structure:** A DT Application is built by single subsystem modules built in multi-physics software and data-driven tools to emulates the system behavior.
- **Trackability:** The system time evolution can be recorded using different model instances as time capsules for the current and previous states of the system.
- **Reprogramability:** DT modules can be reprogramed according to the new system requirements, physical changes, and current system status.

On the other hand, the principal components of a Digital Twin are shown in Fig.2. It can be observed that a DT application is composed of sensors and actuators in the physical system, which collect the system information and apply the control actions over the system. Also, the DT uses the data collected from the system to feed the Multidomain Simulations and perform analytics not only over the DT simulations but also over the real system to produce recommendations and control actions over the system to improve its operation.

### *B. Smart Control Engineering*

SCE begins with the conception of Cognitive Process Control (CPC) proposed by Chen [8] where a process control can be considered cognitive if the controller architecture is aware of the process of vital signs for healthy runs, performs decision making, and health issues alerting using multiple information sources. Also, the control system is capable of learning from past actions and induced errors (resilience), discovering hidden patterns and anomalous behaviors at multiple time scales. So, the process of Cognition, decision, and control is performed integrating a big picture of the system, creating a smart control system, which according to the Smart and Autonomous Systems (S&AS) program of NSF (Natural Sciences Foundation) [17] has the following characteristics:

- **Cognizant:** the system is aware of capabilities and limitations to face the dynamic changes and variability.
- **Taskable:** The system is capable of handle high-level, often vague instructions based on the automated stimulus or human commands.
- **Reflective:** A smart system is able to learn from previous experience to improve its performance.
- **Knowledge Rich:** All the reasoning processes are performed over a diverse body of knowledge from a rich environment and sensor-based information.
- **Ethical:** The smart behavior of a system adheres to a system of societal and legal norms.

So, SCE is a new control framework which combines cognition capabilities with automatic control to introduce intelligence for Industry 4.0 manufacturing processes, supported by IAI and breaking technologies like Cloud computing, Deep learning, Data analytics, or Big data. However, the new capabilities and control paradigm requires realistic virtual representations of the system, making Digital Twin necesary to make possible CPC and SCE. As consequence of DT, a set of enabled capabilities can be leveraged towards SCE applications. Some of these new capabilities are the Smart Control System Design [18], Control Performance Assessment [19], Self Optimizing Control [20], Prognosis and fault detection [21], or parallel control [22].

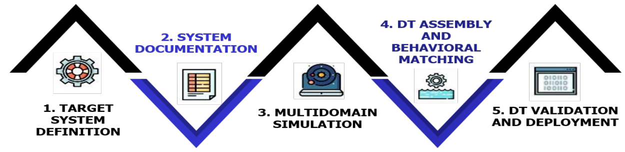


Fig. 3: Digital Twin framework

## III. Development Framework for Digital Twin Applications

Figure 3 presents a framework for the implementation of a DT. The framework is composed of five steps corresponding to the Target System Definition, System Documentation, Multidomain Simulation, DT Assembly and Behavioral Matching, and the DT Evaluation and Deployment.

### *A. Step 1: Target System Definition*

This step recognizes the current status of the physical system to be replicated via Digital Twin with two possible scenarios. The first one is if the physical system is in a conceptual design stage, so building a Digital Twin of the system is a preliminary step for physical sizing as well as emulate its real behavior as closer to reality as possible. In this case, using CAD/CAM tools and design is the best choice to create a real system representation. The second scenario occurs when the physical system is already operative, performing the desired tasks. In this case, the Digital Twin can be also be built using as reference the current system configuration using the system big data, information, and experimental knowledge provided by designers, engineers, and operators. In this paper, the second scenario is applied to the study case.

### *B. Step 2: System Documentation*

Once the target and operating scenario of the system is defined, the next step consists of collect all the available information of the system to create the most accurate representation. This relevant information includes the control algorithms employed (PID, MPC, State Space) and its digital implementation, Sensors and actuators datasheets, troubleshooting and problem records, and system domain knowledge. Notice that most of each subsystem data is available and can be acquired via sensors, virtual metrology, indirect measurements, or state estimators. In addition, the information about the environment where the system performs its tasks is crucial for the correct operation of the Digital Twin like wind speed, direction, environmental temperature, humidity, among other variables relevant for each application.

### *C. Step 3: Multidomain Simulation*

In this step, the definition and configuration of the simulation models for representing the real system behavior are performed. The first task is defining the simulation domains related to the system. It means, defines the physical and constitutive laws that domain the system and select the computational tools to represent it. The domains include thermal, electrical, fluids, or digital components that can be simulated using multiple physics-based simulators like COMSOL, ANSYS, MSC-ADAMS, Matlab Simscape, among other multiphysics software packages. In some cases, the system model also incorporates discrete and algorithmic elements like task scheduling or event-based situations that can be managed using coding. Sometimes, when there is not enough information about a physical model of the system, data-driven models of the system can be employed as a black box to represent that unknown behavior. Once the simulation domains are defined, and each subsystem interaction is built using the corresponding computational simulation tools, which are integrated to reproduce the system behavior.

### *D. Step 4: DT assembly and Behavioral Matching*

In this step, a stable and operative Multidomain Simulation model of the system is available for the Digital Twin realization. Although, this model is operating under ideal conditions as stated before. So, a process called Behavioral Matching (BM) should be performed. It can be defined as a procedure to find the parameters of each subsystem composing the Digital Twin in order to fit its complete system dynamics, coinciding with the real state of the physical system. A description of the Behavioral Matching is presented in Fig.4. As can be observed, the complete Digital Twin model is feed with the input and output data from the real system, including the reference signals for control loops. Based on these data, the Digital Twin is set into an optimization loop, in which the main goal is to determine via optimal searching the unknown parameters that cannot be determined a-priori in each subsystem that compose the Digital Twin. This optimization loop keeps working until the coincidence between the input and output data streams of the system is reached with a certain tolerance rate or after a fixed number of iterations.

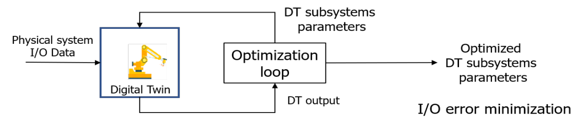


Fig. 4: DT Behavioral Matching

### *E. Step 5: DT Validation and Deployment*

This is the final step on the Digital Twin implementation, performing the validation and simultaneous deployment with the real system. Initially, starting from the Behavioral Matching of the system, the Digital Twin response is calculated for different input/output data sets collected for the system. With this purpose, a supervisory interface has to be designed in order to perform the Digital Twin offline and simultaneous execution of the system. Likewise, through the interface include the Behavioral Matching capabilities, combined with the analytics and fault detection modules, corresponding to the new enabling capabilities for the Digital Twin application.

## IV. Study case: Real-time vision feedback infrared temperature uniformity control

### *A. Step 1: Target System Definition*

The real-time vision feedback infrared temperature uniformity control presented in Fig.5 is employed in this paper as a study case for developing its Digital Twin based on the proposed methodology in Section III. As shown in Fig.5, The system is composed of a Peltier cell (M1) that works as a heating or cooling element, a thermal infrared camera (M2) acting as a temperature feedback sensor running on a Raspberry Pi to perform temperature distribution measurement and control. A LattePanda board (M3) runs Windows 10 64-bits and executes Matlab for local and remote operation. The power applied to the Peltier cell is managed with an Arduino (M4) via Pulse Width Modulation (PWM). A battery (M5) provides the power for the system with four hours of autonomy. In this case, the uniform temperature control system is open and closed-loop stable, employing a PID controller with antwindup.


Fig. 5: DT Study case: real-time vision feedback infrared temperature uniformity control

TABLE I: Brief thermal system documentation

| Component | Features |
|---|---|
| FLIR lepton Thread Infrared thermal Camera | Wavelength: 8 to 14 $\mu$m Resolution: 80x60 pixels Accuracy: $\pm$ 0.5$^oC$ |
| TEC1-12706 Peltier Module | $Q_{max} = 50W$ $\Delta_{Tmax} = 75^oC$ $I_{Max} = 6.4A$ $V_{max} = 16.4V$ |
| MC33926 DC Power Driver | Input: 0-5 V Output: 0-12V Peak Current: 5A |
| Lattepanda board | 5 inch Windows 10 64 bits PC Intel Atom $\mu p$ 4GB of RAM built-in Arduino Leonardo |

### *B. Step 2: System documentation*

The system is composed by the thermal infrared camera, the Peltier module, the control unit, and the power driver [23]–[25]. Table I presents a summary of the critical properties for the mentioned elements. In particular, the Peltier module, the properties differ among manufacturers; therefore, Behavioral Matching is required to determine the right ones. More details about the system implementation and real test performed on the system can be found in [26], [27].

### *C. Step 3: Multidomain Simulation*

The system is divided into four simulation domains shown in Fig.6. The Electrical, composed by the power driver, the Battery, and the semiconductor of the Peltier module. The Thermal domain defined by the heat transfer on the peltier device. The third domain is the fluid given by the airflow pumped into the heat sink. The fourth domain is the Digital Domain, composed by the PID control algorithm and communication interfaces. In this paper, the Electric, Thermal, and Digital domains are replicated in the Digital Twin. Matlab Simulink and Simscape are employed as Multidomain Simulation packages. The complete multiphysics simulation model is presented in Fig.7. The electrical domain is shown in Fig.8. It is composed of the H-bridge to control the power in the Peltier module via

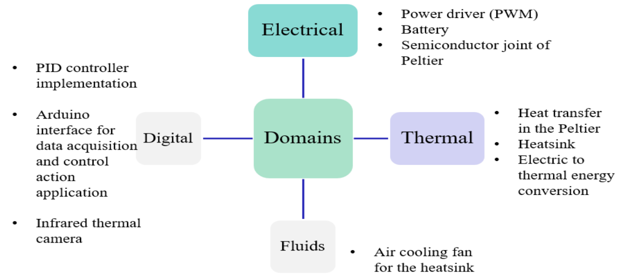


Fig. 6: DT Study case: simulation domains

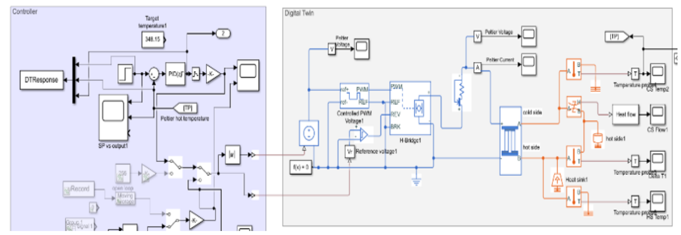

Fig. 7: Assembled DT Multidomain Simulation

PWM with frequency of 500Hz, and current sense threshold of 0.1v. The properties of the H-bridge are defined in Table I. The Thermal domain simulation is presented in Fig.9. As can be observed, the main element on this domain is the Peltier module. Its thermal behavior is modeled using (1)-(3), where $\alpha$ is seebeck coefficient, $R$ electrical resistance, $K$ thermal conductance, $T_A, T_B$ the hot/cold side temperatures, $Q_A, Q_B$ the hot/cold side thermal flow, and $I, V$ Peltier voltage and current. A thermal mass block represents the dynamic change of the heat flow $Q$ in the hot side of the Peltier as (4), where $C$ is the specific heat of the Peltier device and $m$ is the module specific mass. Finally, the heat sink is modeled as a constant temperature source at environment temperature, to producing the differential between A and B.

$$Q_A = \alpha T_A I - \frac{1}{2} I^2 R + K(T_A - T_B) \tag{1}$$

$$Q_B = \alpha T_B I - \frac{1}{2} I^2 R + K(T_B - T_A) \tag{2}$$

$$V = \alpha(T_B - T_A) + IR \tag{3}$$

$$Q = Cm\frac{dT}{dt} \tag{4}$$

The digital domain is shown in Fig.10 and includes the PID controller with anti-wind-up, the reference signal, and the control action to be applied as PWM. For this system, the physical implementation of this domain is performed using Hardware in the Loop simulation with Matlab Simulink. The PID controller gains and configuration are described in detail at [26], [27].

### *D. Step 4: Behavioral Matching*

Behavioral Matching is performed to make the Digital Twin dynamics match with the physical system. For the study case, the thermal domain parameters are the most relevant. So that,

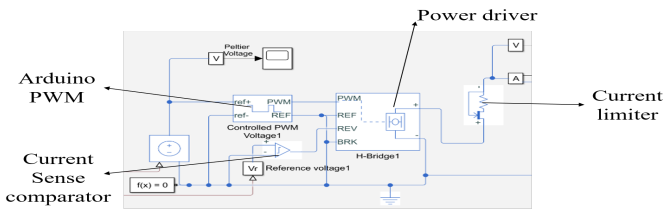


Fig. 8: DT Study case: Electrical domain

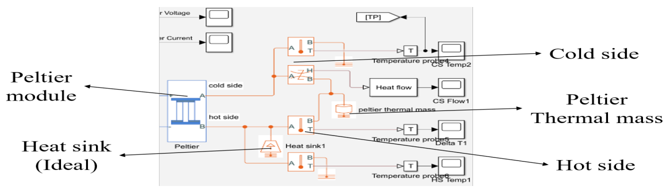


Fig. 9: DT Study case: Thermal domain

finding an accurate $\alpha$, $R$, $K$, and $C$ is required. It can be determined by direct measurement [28], datasheet information [23], or experience. TableII shows the value for these parameters based on the methods described above. To evaluate if Table II parameters make the DT match with the real system, a run of the physical system for a setpoint of $50^{O}C$ is performed, acquiring the control action, Peltier temperature output, and the reference setpoint applied. After that, the Digital Twin Multidomain Simulation is performed with the same reference signal and for each set of parameters in Table.II. Figure.11 shows the Digital Twin responses for each set of parameters compared with the real system. As can be observed, for each set of thermal parameters, the system response looks similar but with different control input behaviors. It means that a Behavioral Matching is required. So that, the Behavioral Matching uses a genetic algorithm (GA), which put the Digital Twin model into an iterative loop looking for the best values $\alpha$,$R$,$K$, and $C$ to match the DT with the real system response. The GA employs the cost function (5), where $J$ is the cost function value, $y_r$ the physical system output, $y_{DT}$ the digital twin output, $u_r$ the control action from the real system, and $u_{DT}$ the control action from Digital Twin, for a period $T$. Also,

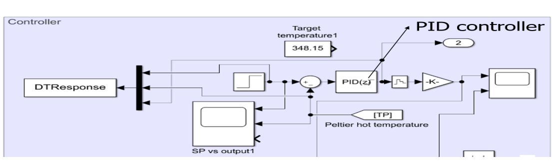


Fig. 10: DT Study case: Digital domain

TABLE II: Peltier Thermal parameters

| Parameter | Datasheet | Measurement | Experience |
|---|---|---|---|
| $\alpha$ | 53 mv | 40 mv | 75 mv |
| $R$ | 1.8 Ω | 6 Ω | 2.90 Ω |
| $K$ | 0.5555 K/W | 0.3333 K/W | 0.3808 K/W |
| C | 15 J/K | 15 J/K | 31.4173 J/K |

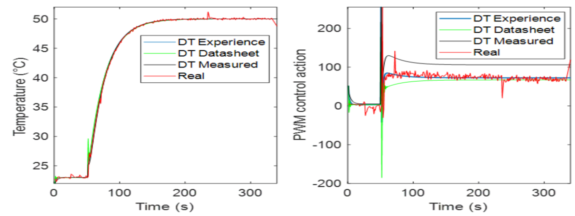


Fig. 11: DT Response for different thermal parameters

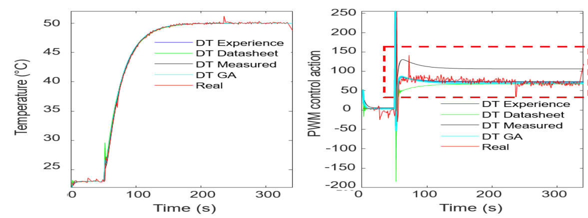


Fig. 12: Behavioral Matching results comparison

the parameters constrains employed are $10mv \leq \alpha \leq 200mv$, $1.8 \leq R \leq 6$, $0.2 \leq K \leq 0.833$, $15 \leq C \leq 30$. The GA is executed for 100 generations with an initial pool of three parents with a mutation probability of 90%, and mutation rate of 2 features per crossing operation. The obtained parameters for the Peltier by the Behavioral Matching are $\alpha = 35.8mv$, $R = 3.35\Omega$ $K = 0.2882K/W$, $C = 15J/K$. Figure.12 shows the response comparison among the Digital Twin for the Peltier parameters in Table II and the obtained with the Behavioral Matching. As can be observed, the optimization-based parameters improve the Digital Twin response, obtaining better fitness than the other parameters sets.

$$min\ J = \int_0^T (y_r - y_{DT})^2 + (u_r - u_D T)^2 dt \qquad (5)$$

### E. Step 5: DT Validation and Deployment

A Matlab supervisory interface is designed to monitor the real system and its Digital Twin as shown in Fig.13. The interface allows interacting with an offline version of the Digital Twin to verify its operation for different setpoints. Likewise, there is a panel for the real-time connection of the system with the Digital Twin. The communication interface among the DT and the system is presented in Fig.14. This interface connects the external computer where the Digital

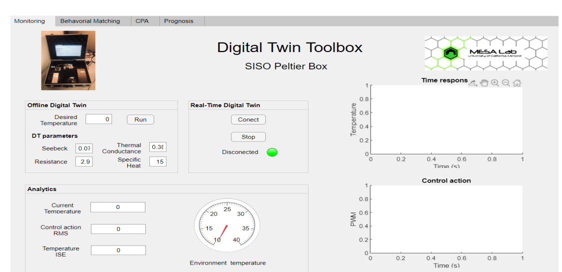


Fig. 13: Digital Twin supervisory interface

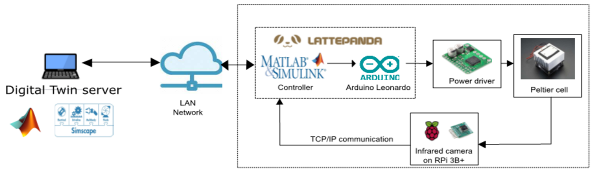


Fig. 14: DT parallel deployment architecture

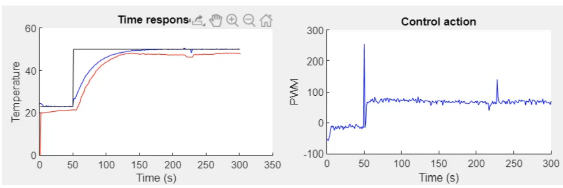


Fig. 15: DT parallel deployment architecture

Twin is executed using a client/server configuration inside a local network with the real thermal system via TCP/IP protocol at 1Hz period. Thus, the Digital Twin is feed in real-time with the real control action, output temperature, environmental temperature among other variables of the physical system to simulate its behavior and measure its difference with the real system. Figure 15 shows the results of the Real-time DT operation (blue line) with the real system (blue line) for a setpoint of $50^oC$. It is observed that the DT replicates the real system behavior under real operation conditions. It is important to notice that the environmental temperature has a significant influence over the Digital Twin performance. For this reason, this variable is measured and incorporated in the Digital Twin modeling and execution. A video recording of the Digital Twin interface operation can be visualized in https://youtu.be/acXTNmcCIYs.

## V. Conclusions and Future Works

This paper presented a methodological framework to develop a Digital Twin for a closed-loop system real-time vision feedback infrared temperature uniformity control towards the implementation of SCE. This framework employs five steps, Target System Definition, system documentation, Multidomain Simulation, DT assembly and Behavioral Matching, and the DT evaluation and deployment. The obtained results show that the Digital Twin represents the system behavior operating in parallel and offline modes. Thus, the Digital Twin of the temperature uniformity control system can incorporate the enabling technologies for SCE. As future works, the introduction of enabling technologies like fault detection and prognosis along with the Digital Twin is proposed.

## References

[1] C. Strategies, "Artificial Intelligence for Industrial Applications," tech. rep., CloudPulse Strategies, 2017.

[2] J. Zhang and J. Cao, "Fault diagnosis for closed loop nonlinear system using generalized frequency response functions and least square support vector machine," *1st International Conference on Industrial Artificial Intelligence, IAI 2019*, vol. 2, no. 2, pp. 2–5, 2019.

[3] G. Li, K. Ota, M. Dong, J. Wu, and J. Li, "DeSVig: Decentralized Swift Vigilance against Adversarial Attacks in Industrial Artificial Intelligence Systems," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 5, pp. 3267–3277, 2020.

[4] M. Hao, H. Li, X. Luo, G. Xu, H. Yang, and S. Liu, "Efficient and Privacy-enhanced Federated Learning for Industrial Artificial Intelligence," *IEEE Transactions on Industrial Informatics*, vol. 3203, no. c, pp. 1–1, 2019.

[5] Y. Mao, W. Zhang, M. Wang, and D. Guo, "Driving simulator data based driver behavior ethogram establishment," *1st International Conference on Industrial Artificial Intelligence, IAI 2019*, 2019.

[6] Y. Dong and X. Li, "Dsmt-based fusion system for fault detection and identification in industrial process," *1st International Conference on Industrial Artificial Intelligence, IAI 2019*, 2019.

[7] Y. Li, Z. Lu, F. Zhou, B. Dong, K. Liu, L. Guangjun, and Y. Li, "Decentralized active fault tolerant control for modular and reconfigurable robot with torque sensor," *1st International Conference on Industrial Artificial Intelligence, IAI 2019*, 2019.

[8] YangQuan Chen, "Cognitive Process Control," 2012.

[9] D. Guivarch, E. Mermoz, Y. Marino, and M. Sartor, "Creation of helicopter dynamic systems digital twin using multibody simulations," *CIRP Annals*, vol. 68, no. 1, pp. 133–136, 2019.

[10] X. Xie, A. K. Parlikad, and R. S. Puri, "Approach for Demand Forecasting within Power Grid Digital Twins," *2019 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, pp. 1–6.

[11] W. Luo, T. Hu, W. Zhu, and F. Tao, "Digital twin modeling method for CNC machine tool," *ICNSC 2018 - 15th IEEE International Conference on Networking, Sensing and Control*, no. 51405270, pp. 1–4, 2018.

[12] F. Zhu, Z. Li, S. Chen, and G. Xiong, "Parallel Transportation Management and Control System and Its Applications in Building Smart Cities," *IEEE Transactions on Intelligent Transportation Systems*, vol. 17, no. 6, pp. 1576–1585, 2016.

[13] F. N. Claessen, F. Claessen, W. V. Sark, E. Worrell, V. Bakker, A. Molderink, H. Toersche, B. Claessens, M. Hommelberg, and U. E. Technology, *Smart grid control*. 2012.

[14] W. Danilczyk, Y. Sun, and H. He, "ANGEL: An Intelligent Digital Twin Framework for Microgrid Security," *51st North American Power Symposium, NAPS 2019*, 2019.

[15] M. Farsi, A. Daneshkhah, A. Hosseinian-Far, and H. Jahankhani, *Internet of Things Digital Twin Technologies and Smart Cities*. Springer, 2020.

[16] F. Tao and Q. Qi, "Make more digital twins," *Nature*, vol. 573, no. 7775, pp. 490–491, 2019.

[17] NSF, "Smart and Autonomous Systems (S&AS)," 2018.

[18] D. Xue and Y. Chen, *Modeling, Analysis and Design of Control Systems in MATLAB and Simulink*. WORLD SCIENTIFIC, 2014.

[19] P. D. Domański, *Control Performance Assessment: Theoretical Analyses and Industrial Practice*. Studies in Systems, Decision and Control, Springer International Publishing, 2019.

[20] S. Skogestad, "Plantwide control: the search for the self-optimizing control structure," *Journal of Process Control*, vol. 10, no. 5, pp. 487–507, 2000.

[21] R. Isermann, *Fault-Diagnosis Systems An Introduction from Fault Detection to Fault Tolerance*, vol. 1. Springer-Verlag New York, Inc., 2006.

[22] M. Kang and F. Y. Wang, "From parallel plants to smart plants: Intelligent control and management for plant growth," *IEEE/CAA Journal of Automatica Sinica*, vol. 4, no. 2, pp. 161–166, 2017.

[23] H. Ltd, "Thermoelectric Cooler TEC1-12706 datasheet," 2010.

[24] F. S. Inc., "FLIR LEPTON Engineering Datasheet," 2018.

[25] DFROBOT, "Lattepanda Documentation," 2018.

[26] J. Viola, P. Oziablo, and Y. Q. Chen, "An Experimental Networked Control System with Fractional Order Delay Dynamics," *2019 IEEE 7th International Conference on Control, Mechatronics and Automation, ICCMA 2019*, pp. 226–231, 2019.

[27] J. Viola, A. Radici, S. Dehghan, and Y. Chen, "Low-cost real-time vision platform for spatial temperature control research education developments," in *Proceedings of the ASME Design Engineering Technical Conference*, vol. 9, 2019.

[28] V. I. Kubov, Y. Y. Dymytrov, and R. M. Kubova, "Ltspice-model of thermoelectric peltier-seebeck element," in *2016 IEEE 36th International Conference on Electronics and Nanotechnology (ELNANO)*, pp. 47–51, 2016.